ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

path to success

PHYSICS

JEE (MAIN+ADVANCED)

NURTURE COURSE



Exercise

Work, Power & Energy

(English Medium)

EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Work

1. A force of magnitude of 30 N acting along $\hat{i}+\hat{j}+\hat{k}$, displaces a particle from point (2, 4, 1) to (3, 5, 2). The work done during this displacement is :-

(A) 90 J (B) 30 J (C) $30\sqrt{3}$ J (D) $30/\sqrt{3}$ J.

WE0038

2. A spring of force constant 800 N/m has an extension of 5 cm. The work done in extending it from 5cm to 15 cm is- **[AIEEE - 2002]**

(A) 16 J (B) 8 J (C) 32 J (D) 24 J

WE0039

3. A rope is used to lower vertically a block of mass M by a distance x with a constant downward acceleration g/2. The work done by the rope on the block is :

(A) Mgx (B) ½ Mgx² (C) – ½ Mgx (D) Mgx²

WE0040

4. A bucket full of water is drawn up by a person. In this case the work done by the gravitational force is:

(A) negative because the force and displacement are in opposite directions.

(B) positive because the force and displacement are in the same direction.

(C) negative because the force and displacement are in the same direction.

(D) positive because the force and displacement are in opposite directions.

WE0154

5. A ball is thrown vertically upwards from the ground. Work done by air resistance during its time of flight is:

(A) positive during ascent and negative during descent.

(B) positive during ascent and descent.

(C) negative during ascent and positive during descent.

(D) negative during ascent and descent.

WE0155

6. A force of 1200 N acting on a stone by means of a rope slides the stone through a distance of 10m in a direction inclined at 60° to the force. The work done by the force is:

(A) $6000\sqrt{3}$ J (B) 6000J (C) 12000J (D) 8000J

WE0156

7. A man weighing 80 kg climbs a staircase carrying a 20 kg load. The staircase has 40 steps, each of 25cm height. If he takes 20 seconds to climb, the work done is:

(A) 9800J (B) 490 J (C) 98×10^5 J (D) 7840J

WE0157

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8. **This question has Statement-1 and Statement-2. Of the four choices given after the statements, choose the one that best describes the two statements.** **[AIEEE-2012]**

If two springs S_1 and S_2 of force constants k_1 and k_2, respectively, are stretched by the same force, it is found that more work is done on spring S_1 than on spring S_2.

Statement-1: If stretched by the same amount, work done on S_1, will be more than that on S_2

Statement-2 : $k_1 < k_2$

(A) Statement-1 is true, Statement-2 is true and Statement-2 is not the correct explanation of Statement-1.

(B) Statement-1 is false, Statement-2 is true

(C) Statement-1 is true, Statement-2 is false

(D) Statement-1 is true, Statement-2 is true and Statement-2 is the correct explanation of statement-1.

WE0089

9. A force F is applied on a lawn mover at an angle of 60° with the horizontal. If it moves through a distance x, the work done by the force is: **WE0192**

(A) Fx/2 (B) F/2x (C) 2Fx (D) 2x/F

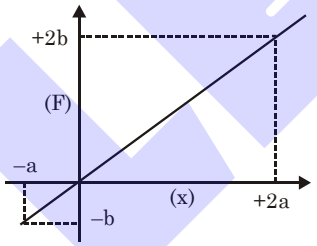
10. A force F = (2+x) N acts on a particle in x - direction where 'x' is in metre. The work done by this force during a displacement from x = 1 m to x = 2 m is : **WE0193**

(A) 2 J (B) 3.5 J (C) 4.5 J (D) 5 J

11. A body of mass 5 kg is moved up over 10 m along the line of greatest slope of a smooth inclined plane of inclination 30° with the horizontal. If g = 10 m/s^2 , the work done will be :

(A) 500 J (B) 2500 J (C) 250 J (D) 25 J **WE0158**

12. A force F acting on a particle varies with the position x as shown in the graph. Find the work done by the force in displacing the particle from x = –a to x = +2a **WE0194**



(A) $\frac{3ab}{2}$ (B) $\frac{4ab}{2}$ (C) $\frac{2}{3ab}$ (D) $\frac{2}{4ab}$

13. A body is displaced from (0,0) to (1m, 1m) along the path x = y by a force $F = (x^2\hat{j} + y\hat{i})N$. The work done by this force will be: **WE0195**

(A) $\frac{4}{3}$J (B) $\frac{5}{6}$J (C) $\frac{3}{2}$J (D) $\frac{7}{5}$J

Kinetic Energy

14. A man who is running has half the kinetic energy of the boy of half his mass. The man speeds up by 1 m/s and then has the same kinetic energy as the boy. The original speed of the man was

(A) $\sqrt{2}$ m/s (B) $(\sqrt{2} - 1)$ m/s (C) 2 m/s (D) $(\sqrt{2} + 1)$ m/s

WE0045

15. A mass of 5 kg is moving along a circular path of radius 1 m. If the mass moves with 300 revolutions per minute, its kinetic energy would be

(A) 250 π^2 joule (B) 100π^2 joule (C) 5π^2 joule (D) 0 joule

WE0046

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16. For the same kinetic energy, the momentum shall be maximum for which of the following particle?
(A) Electron (B) Proton (C) Deuteron (D) Alpha particle
WE0196

17. Consider the following statements **WE0197**
(1)) Linear momentum of a system of particles is zero
(2) Kinetic energy of a system of particles is zero then
(A) 1 does not imply 2 & 2 does not imply 1.
(B) 1 implies 2 and 2 does not imply 1.
(C) 1 does not imply 2 but 2 implies 1.
(D) 1 implies 2 and 2 implies 1.

Work energy theorem

18. The KE of a freely falling body: **WE0198**
(A) is directly proportional to height of its fall.
(B) is inversely proportional to height of its fall.
(C) is directly proportional to square of time of its fall.
(D) A and C are true.

19. A body starts from rest and is acted on by a constant force. The ratio of kinetic energy gained by it in the first five seconds to that gained in the next five seconds is: **WE0199**
(A) 2 : 1 (B) 1 : 1 (C) 3 : 1 (D) 1 : 3

20. The speed of a car changes from 0 to 5 ms^{-1} in the first phase and from 5 ms^{-1} to 10 ms^{-1} in the second phase and from 10 ms^{-1} to 15 ms^{-1} during the third phase. In which phase the increase in kinetic energy is more?
(A) first phase (B) second phase
(C) third phase (D) same in all the three phases **WE0160**

21. A vertically projected body attains the maximum height in 6s. The ratio of kinetic energy at the end of 3rd second to decrease in kinetic energy in the next three seconds is: **WE0200**
(A) 1: 1 (B) 1 : 3 (C) 3 : 1 (D) 9 : 1

22. At time t = 0s particle starts moving along the x-axis. If its kinetic energy increases uniformly with time 't', the net force acting on it must be proportional to :- **[AIEEE-2011]**
(A) $\sqrt{t}$ (B) constant (C) t (D) $\frac{1}{\sqrt{t}}$ **WE0088**

23. A particle moves on a rough horizontal ground with some initial velocity say v_0. If 3/4 of its kinetic energy is lost in friction in time t_0 then coefficient of friction between the particle and the ground is :-
(A) $\frac{v_0}{2gt_0}$ (B) $\frac{v_0}{4gt_0}$ (C) $\frac{3v_0}{4gt_0}$ (D) $\frac{v_0}{gt_0}$ **WE0049**

24. A block of mass m is hung vertically from an elastic thread of force constant mg/a. Initially the thread was at its natural length and the block is allowed to fall freely. The kinetic energy of the block when it passes through the equilibrium position will be :
(A) mga (B) mga/2 (C) zero (D) 2mga **WE0051**

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25. A lorry and a car moving with the same KE are brought to rest by applying the same retarding force. Then

(A) lorry will come to rest in a shorter distance.

(B) car will come to rest in a shorter distance.

(C) both come to rest in same distance.

(D) any of above. **WE0161**

26. A shot is fired at 30° with the vertical from a point on the ground with kinetic energy K. If air resistance is ignored, the kinetic energy at the top of the trajectory is:

(A) 3K/4 (B) K/2 (C) K (D) K/4 **WE0201**

27. A body of mass 2 kg is projected with an initial velocity of 5 ms^{-1} along a rough horizontal table. The work done on the body by the frictional forces before it is brought to rest is:

(A) 250 J (B) 25 J (C) –250 J (D) –25 J **WE0162**

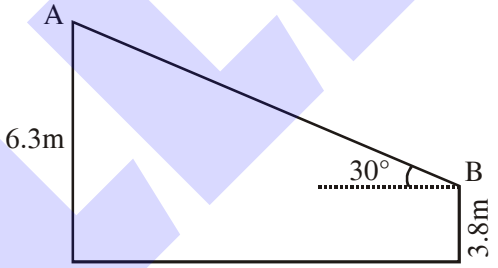
28. By applying the brakes without causing a skid, the driver of a car is able to stop his car with in a distance of 5 m, if it is going at 36 kmph. If the car were going at 72 kmph, using the same brakes, he can stop the car over a distance of:

(A) 10 m (B) 2.5 m (C) 20 m (D) 40 m **WE0163**

29. A bullet fired into a trunk of a tree loses 1/4 of its kinetic energy in travelling a distance of 5 cm. Before stopping it travels a further distance of:

(A) 150 cm (B) 1.5 cm (C) 1.25 cm (D) 15 cm **WE0164**

30. AB is a frictionless inclined surface making an angle of 300 with horizontal. A is 6.3 m above the ground while B is 3.8 m above the ground. A block slides down from A, initially starting from rest. Its velocity on reaching B is:



(A) 7 m s^{-1} (B) 14 m s^{-1} (C) 7.4 m s^{-1} (D) 4.9 m s^{-1} **WE0202**

31. The workdone to accelerate a body from 30 m/s to 60 m/s is three times the work done to accelerate it from 10 m/s to 'v' . The value of 'v' in m/s is:

(A) 30 (B) $20\sqrt{2}$ (C) $30\sqrt{3}$ (D) $10\sqrt{10}$ **WE0203**

32. A block of mass 4 kg is initially at rest on a horizontal frictionless surface. A horizontal force $\vec{F} = (3+x)\hat{i}$ newtons acts on it, when the block is at x=0. The maximum kinetic energy of the block between x = 0 and x = 2m is

(A) 6J (B) 8J (C) 9J (D) 10J **WE0166**

33. A block of mass 4 kg slides on a horizontal frictionless surface with a speed of 2m/s. It is brought to rest in compressing a spring in its path. If the force constant of the spring is 400 N/m, by how much the spring will be compressed

(A) 2×10^{-2}m (B) 0.2m (C) 20 m (D) 200m **WE0167**

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34. A block of mass 2 kg is on a smooth horizontal surface. A light spring of force constant 800 N/ m has one end rigidly attached to a vertical wall and lying on that horizontal surface. Now the block is moved towards the wall compressing the spring over a distance of 5 cm and then suddenly released. By the time the spring regains its natural length and looses contact with the block, the velocity acquired by the block will be:

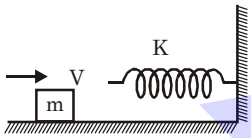
(A) 200m/s (B) 100 m/s (C) 2m/s (D) 1m/s **WE0204**

35. A bullet of mass 10gm is fired horizontally with a velocity 1000ms^{-1} from a riffle situated at a height 50 m above the ground. If the bullet reaches the ground with a velocity 500ms^{-1}, the work done against air resistance in the trajectory of the bullet is (in joule) (g = 10ms^{-2})

(A) 5005 (B) 3755 (C) 3750 (D) 17.5

WE0168

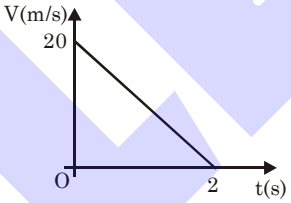
36. A block of mass m = 25kg on a smooth horizontal surface with a velocity $\vec{v}$ = 3ms^{-1} meets the spring of spring constant k = 100 N/m fixed at one end as shown in figure. The maximum compression of the spring and velocity of block as it returns to the original position respectively are :



(A) 1.5m, –3 m/s (B) 1.5m, 0 m/s (C) 1.0m, 3 m/s (D) 0.5m, 2 m/s

WE0169

37. Velocity -time graph of a particle of mass 2kg moving in a straight line is as shown in the figure. Work done by all the forces on the particle is:



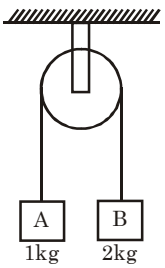
(A) 400J (B) –400J (C) –200J (D) 200J

WE0170

38. A force F=Ay 2+By+C acts on a body at rest in the Y-direction. The kinetic energy of the body during a displacement y = - a to y = a is: **WE0205**

(A) $\frac{2Aa^3}{3}$ (B) $\frac{2Aa^3}{3}+2ca$ (C) $\frac{2Aa^3}{3}+\frac{Ba^3}{2}+ca$ (D) $\frac{2Aa^3}{3}+\frac{Ba^3}{2}$

39. In the arrangement shown in figure, string is light and inextensible and friction is absent every where. The speed of both blocks after the block 'A' has ascend a height of 1m will be (g = 10 m/s^2)

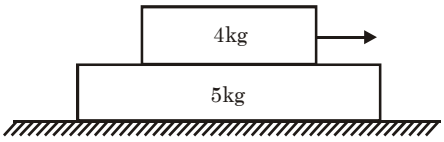


(A) 2m/s (B) 2.58 m/s (C) 3m/s (D) 3.58 m/s

WE0171

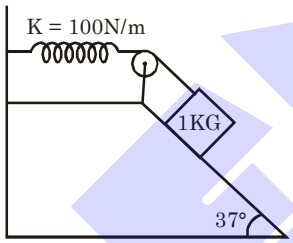
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40. A large slab of mass 5 kg lies on a smooth horizontal surface, with a block of mass 4 kg lying on the top of it. The coefficient of friction between the block and the slab is 0.25. If the block is pulled horizontally by a force of F = 6 N, then the work done by the force of friction on the slab, between the instants t = 2s to t = 3s is ($g = 10\ ms^{-2}$) **WE0207**



(A) 2.4 J (B) 5.55 J (C) 4.44J (D) 10 J

41. A 1 kg block situated on a rough inclined plane is connected to spring of a spring constant 100 Nm-1 as shown in fig. The block is released from rest with the spring in the unstretched position. The block moves 10 cm down the incline before coming to rest. Find the coefficient of friction between the block and the incline. Assume that the spring has a negligible mass and the pulley is frictionless. ($g = 10 m/s^2$)



(A) 0.125 (B) 1.25 (C) 5.2 (D) 4.5

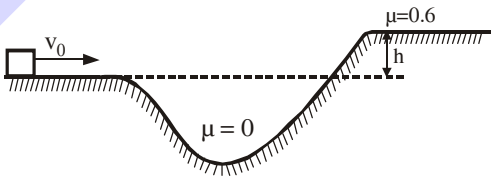
WE0172

42. A body of mass 0.5 kg travels in a straight line with velocity $v = ax^{3/2}$ where $a = 5\ m^{-1/2}s^{-1}$. The work done by the net force during its displacement from $x = 0$ to $x = 2$ m is

(A) 1.5 J (B) 50 J (C) 10 J (D) 100 J

WE0044

43. In the figure, a block slides along a track from one level to a higher level, by moving through an intermediate valley. The track is frictionless untill the block reaches the higher level. There a frictional force stops the block in a distance d. The block's initial speed v_0 is 6 m/s, the height difference h is 1.1m and the coefficient of kinetic friction μ is 0.6. The value of d is



(A) 1.17 m (B) 1.71 m (C) 7.11 m (D) 11.7 m

WE0047

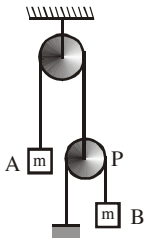
44. In a shotput event an athlete throws the shotput of mass 10 kg with an initial speed of 1m/s at 45° from a height 1.5 m above ground. Assuming air resistance to be negligible and acceleration due to gravity to be 10 m/s^2, the kinetic energy of the shotput when it just reaches the ground will be

(A) 2.5 J (B) 5.0 J (C) 52.5 J (D) 155.0 J

WE0048

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45. In the figure shown, the system is released from rest. Find the velocity of block A when block B has fallen a distance 'ℓ'. Assume all pulleys to be massless and frictionless.

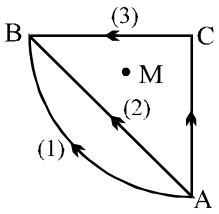


(A) $\sqrt{\frac{g\ell}{5}}$ (B) $\sqrt{g\ell}$ (C) $\sqrt{5g\ell}$ (D) None of these

WE0050

Potential energy

46. In a region of only gravitational field of mass 'M' a particle is shifted from A to B via three different paths in the figure. The work done in different paths are W_1, W_2, W_3 respectively then **[IIT-JEE (Scr.)'2003]**

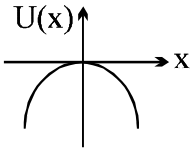


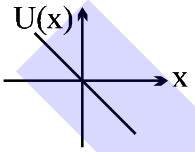
(A) $W_1 = W_2 = W_3$
(B) $W_1 = W_2 > W_3$
(C) $W_1 > W_2 > W_3$
(D) $W_1 < W_2 < W_3$

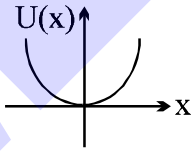
WE0043

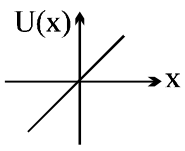
47. A particle is placed at the origin and a force F = kx is acting on it (where k is a positive constant). If U(0) = 0, the graph of U(x) versus x will be (where U is the potential energy function)

[IIT-JEE' 2004(Scr)]

(A) 

(B) 

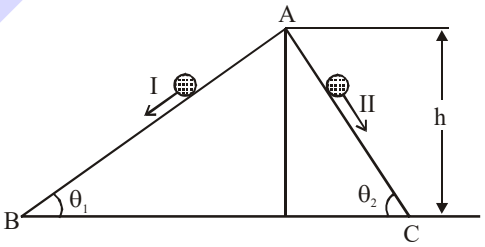
(C) 

(D) 

WE0059

48. When a conservative force does positive work on a body
(A) the potential energy increases (B) the potential energy decreases
(C) total energy increases (D) total energy decreases **WE0060**

49. Two inclined frictionless tracks, one gradual and the other steep meet at A from where two stones are allowed to slide down from rest, one on each track as shown in figure. Which of the following statement is correct?



(A) Both the stones reach the bottom at the same time but not with the same speed.
(B) Both the stones reach the bottom with the same speed and stone I reaches the bottom earlier than stone II.
(C) Both the stones reach the bottom with the same speed and stone II reaches the bottom earlier than stone I.
(D) Both the stones reach the bottom at different times and with different speeds.

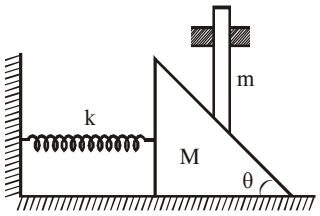
WE0057

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50. A wedge of mass M fitted with a spring of stiffness 'k' is kept on a smooth horizontal surface. A rod of mass m is kept on the wedge as shown in the figure. System is in equilibrium. Assuming that all surfaces are smooth, the potential energy stored in the spring is:

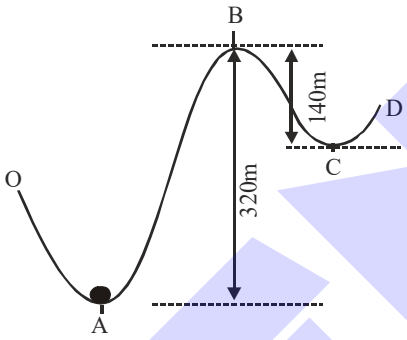
(A) $\frac{mg^2\tan^2\theta}{2k}$ (B) $\frac{m^2g\tan^2\theta}{2k}$

(C) $\frac{m^2g^2\tan^2\theta}{2k}$ (D) $\frac{m^2g^2\tan^2\theta}{k}$



WE0061

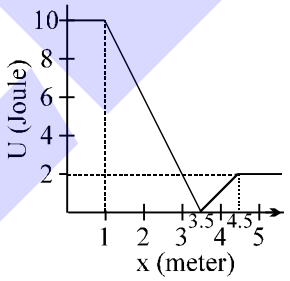
51. Track OABCD (as shown is figure) is smooth and fixed in vertical plane. What minimum speed has to be given to a particle lying at point A, so that it can reach point C?



(A) 60 m/s (B) 100 m/s (C) 70 m/s (D) 80 m/s

WE0058

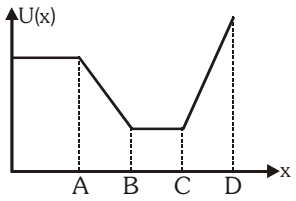
52. A body with mass 2 kg moves in one direction in the presence of a force which is described by the potential energy graph. If the body is released from rest at x = 2m, then its speed when it crosses x = 5 m is



(A) zero (B) 1 ms^{-1} (C) 2 ms^{-1} (D) 3 ms^{-1}

WE0062

53. As a particle moves along the x–axis it is acted upon by a conservative force. The potential energy is shown below as a function of the coordinate x of the particle. Rank the labelled regions according to the magnitude of the force, least to greatest.

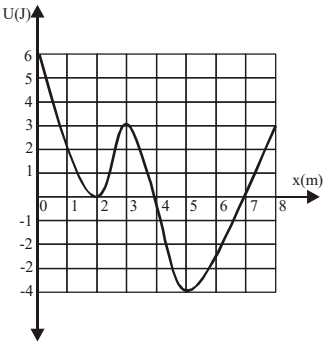


(A) AB, BC, CD (B) AB, CD, BC (C) BC, CD, AB (D) BC, AB, CD

WE0064

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54. Potential energy curve U of a particle as function of the position of a particle is shown. The particle has total mechanical energy E of 3.0 joules.



(A) It can never be present at x = 0 m.
(B) It can never be present at x = 5 m
(C) At x = 2 its kinetic energy is 0 J
(D) At x = 1 its kinetic energy 3 J

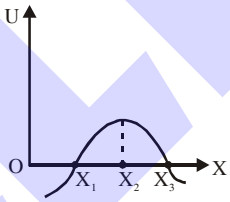
WE0065

55. The total mechanical energy of a particle is E. The speed of the particle at $x = \left(\frac{2E}{K}\right)^{1/2}$ is $\left(\frac{2E}{m}\right)^{1/2}$.

Find the potential energy of the particle at x :

(A) zero (B) $\frac{1}{2}Kx^2$ (C) $\frac{1}{4}Kx^2$ (D) $\frac{2}{5}Kx^2$ **WE0208**

56. In the figure shown the potential energy (U) of a particle is plotted against its position 'x' from origin. Then which of the following statement is correct. A particle at :



(A) x_1 is in stable equilibrium
(B) x_2 is in stable equilibrium
(C) x_3 is in stable equilibrium
(D) None of these

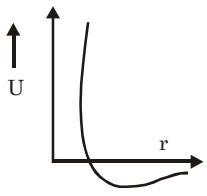
WE0063

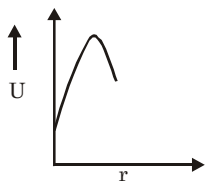
57. The potential energy function for the force between two atoms in a diatomic molecule is approximately given by $U(x) = \frac{a}{x^{12}} - \frac{b}{x^6}$, where a and b are constant and x is the distance between the atoms. if the dissociation energy of the molecule is $D = [U(x = \infty) - U_{\text{at equilibrium}}]$, D is : **[AIEEE-2010]**

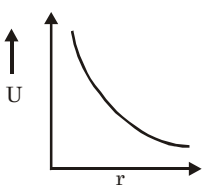
(A) $\frac{b^2}{6a}$ (B) $\frac{b^2}{2a}$ (C) $\frac{b^2}{12a}$ (D) $\frac{b^2}{4a}$

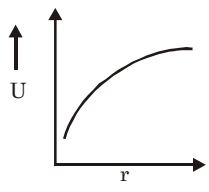
WE0087

58. These diagrams represent the potential energy U of a diatomic molecule as a function of the inter-atomic distance r. The diagram corresponds to stable molecule found in nature is: **WE0209**

(A) 

(B) 

(C) 

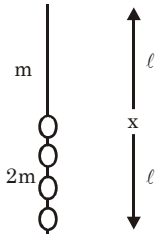
(D) 

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59. A tank of size 10 m × 10 m × 10 m is full of water and built on the ground. If g = 10 ms^{-2}, the potential energy of the water in the tank is :

(A) 5 × 10^{7}J (B)1 × 10^8 J (C)5 × 10^4 J (D) 5 × 10^5 J **WE0174**

60. A rope of length ℓ and mass 'm' is connected to a chain of length l and mass 2m and hung verticlly as shown. what is the change in gravitational potential energy if the system is inverted and hung from same point. **WE0210**



(A) mgℓ (B) 4mgℓ (C) 3mgℓ (D) 2mgℓ

Power

61. A motor boat is going in a river with a velocity $\vec{V} = (4\hat{i} - 2\hat{j} + \hat{k})ms^{-1}$. If the resisting force due to stream is $\vec{F} = (5\hat{i} - 10\hat{j} + 6\hat{k})N$, then the power of the motor boat is: **WE0211**

(A) 100 W (B) 50 W (C) 46 W (D) 23 W

62. A man carries a load of 50 kg through a height of 40 m in 25 seconds. If the power of the man is 1568 W, his mass is: **WE0212**

(A) 5 kg (B) 1000 kg (C) 200 kg (D) 50 kg

63. An electric motor creates a tension of 4500 newton in a hoisting cable and reels it at the rate of 2m/s. What is the power of the motor? **WE0213**

(A) 15 kW (B) 9 kW (C) 225 W (D) 9000 kW

64. A car weighing 1000 kg is going up an incline with a slope of 2 in 25 at a steady speed of 18 kmph. If g= 10 m/s^2, the power of its engine is:

(A) 4 kW (B) 50 kW (C) 625 kW (D) 25 kW **WE0175**

65. An electric motor operates with an efficiency of 90%. A pump operated by the motor has an efficiency of 80%. The overall efficiency of the system is:

(A) 85% (B) 100% (C) 72% (D) 60% **WE0214**

66. A machine rated as 150W, changes the velocity of a 10kg mass from 4m/s to 10m/s in 4s. The efficiency of the machine is nearly.

(A) 70% (B) 30% (C) 50% (D) 40% **WE0215**

67. A pump is required to lift 600 kg of water per minute from a well 25m deep and to eject it with a speed of 50ms^{-1}. The power required to perform the above task is:

(A) 10 kW (B) 15 kW (C) 20 kW (D) 25 kW **WE0176**

68. A car drives along a straight level frictionless road by an engine delivering constant power. Then velocity is directly proportional to

(A) t (B) $\frac{1}{\sqrt{t}}$ (C) $\sqrt{t}$ (D) t^2 **WE0177**

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Motion in vertical circle

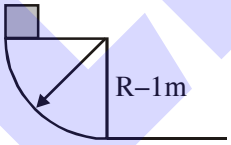
69. A vehicle is travelling along concave road then along convex road of same radius of curvatures at uniform speed. If the normal reactions on the vehicle as it crosses the lowest point of concave surface, highest point of convex surface are 1.5×10^4N, 3×10^3N respectively, then the mass of vehicle is($g = 10\ m/s^{-2}$)
($g = 10\ m/s^{-2}$)
(A) 400 kg (B) 450 kg (C) 800 kg (D) 900kg
WE0178

70. A simple pendulum of length 'l' carries a bob of mass 'm'. If the breaking strength of the string is 2 mg, the maximum angular amplitude from the vertical can be:
(A) 0° (B) 30° (C) 60° (D) 90°
WE0179

71. A point size mass 100 gm is rotated in a vertical circle using a cord of length 20cm. When the string makes an angle 60° with the vertical , the speed of the mass is 1.5m/s. The tangential acceleration of the mass in that position is (in m/s^2)
(A) 4.9 (B) $4.9\sqrt{2}$ (C) $4.9\sqrt{3}$ (D) 9.8 **WE0216**

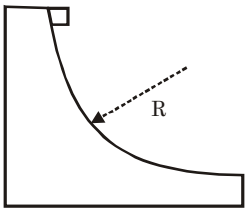
72. The length of a simple pendulum is 1 m. The bob is given a velocity 7 m/s in horizontal direction from mean position. During upward motion of bob, if the string breaks when the bob is horizontal, then the maximum vertical height of ascent of bob from rest position is :
(A) 2.5 m (B) 2 m (C) 3 m (D) 3.5 m
WE0180

73. A block of mass of 1kg slides down a curved track that is one -quadrant of circle of radius 1m. Its speed at the bottom is 2m/s. The workdone by the frictional force is **WE0217**



(A) 8J (B) –8J (C) 4J (D) –4J

74. A body slides down a fixed curved track that is one quadrant of a circle of radius R, as in the figure. If there is no friction and the body starts from rest, its speed at the bottom of the track is: **WE0218**



(A) 5gR (B) $\sqrt{5gR}$ (C) $\sqrt{2gR}$ (D) $\sqrt{gR}$

75. A body of mass m is rotated in a vertical circle with help of light string such that velocity of body at a point is equal to critical velocity at that point. If T_1, T_2 be the tensions in the string when the body is crossing the highest and the lowest positions then the following relation is correct **WE0219**
(A) $T_2 - T_1 = 6mg$ (B) $T_2 - T_1 = 4mg$ (C) $T_2 - T_1 = 3mg$ (D) $T_2 - T_1 = 2mg$

76. A car is travelling along a flyover bridge which is a part of vertical circle of radius 10m. At the highest point of it the normal reaction on the car is half of its weight, the speed of car is: **WE0220**
(A) 7m/s (B) 10m/s (C) 14m/s (D) 20m/s

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77. A uniform cylinder of radius 'r' length 'L' and mass 'm' is lying on the ground with the curved surface touching the ground. If it is to be oriented on the ground with the flat circular end in contact with the ground, the work to be done is **WE0221**

(A) mg[(L/2) – r] (B) mL[(g/2) – r] (C) mr(gL – 1) (D) mgLr

78. A ball of mass 0.6kg attached to a light inextensible string rotates in a vertical circle of radius 0.75m such that it has speed of $5ms^{-1}$ when the string is horizontal. Tension in the string when it is horizontal on other side is ($g = 10\ m/s^2$) [2007M]

(A) 30N (B) 26N (C) 20N (D) 6N

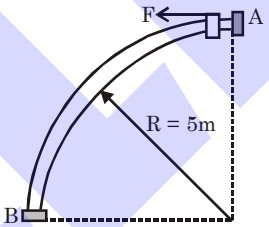
WE0181

79. A body of mass m is rotated in a vertical circle of radius R by means of light string. If the velocity of body is $\sqrt{gR}$ while it is crossing highest point of vertical circle then the tension in the string at that instant is:

(A) 2mg (B) mg (C) $\frac{mg}{2}$ (D) Zero

WE0182

80. A bead of mass $\frac{1}{2}$kg starts from rest from "A" to move in a vertical plane along a smooth fixed quarter ring of radius 5m, under the action of a constant horizontal force F = 5 N as shown. The speed of bead as it reaches point "B" is: **WE0222**



(A) 14.14 m/s (B) 7.07 m/s (C) 5 m/s (D) 25 m/s

MULTIPLE CORRECT TYPE QUESTIONS

Work energy theorem

81. A man is standing on a plank which is placed on smooth horizontal surface. There is sufficienet friction between feet of man and plank. Now man starts running over plank, correct statement is/are

(A) Work done by friction on man with respect to ground is nagative.

(B) Work done by friction on man with respect to ground is positive.

(C) Work done by friction on plank with respect to ground is positive.

(D) Work done by friction on man with respect to plank is zero. **WE0183**

82. Which of the following statements is TRUE for a system comprising of two bodies in contact exerting frictional force on each other :

(A) total work done by static friction on whole system is always zero.

(B) work done by static friction on a body is always zero

(C) work done by kinetic friction on a body is always negative

(D) total work done by internal kinetic friction on whole system is always negative

WE0067

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83. A particle of mass *m* is at rest in a train moving with constant velocity with respect to ground. Now the particle is accelerated by a constant force F_0 acting along the direction of motion of train for time t_0. A girl in the train and a boy on the ground measure the work done by this force. Which of the following are **INCORRECT**?

(A) Both will measure the same work

(B) Boy will measure higher value than the girl

(C) Girl will measure higher value than the boy

(D) Data are insufficient for the measurement of work done by the force F_0 **WE0068**

Potential energy

84. A particle of mass m = 1 kg lying on x–axis experiences a force given by law F=x(3x–2) Newton, where x is the x–coordinate of the particle in meters. The points on x–axis where the particle is in equilibrium are :

(A) x = 0 (B) x = 1/3 (C) x = 2/3 (D) x = 1

WE0081

85. If one of the forces acting on a particle is conservative then :

(A) Work done by this force is zero when the particle moves exactly once around any closed path.

(B) Work done by this force equals the change in the kinetic energy of the particle.

(C) It obeys Newton's second law.

(D) Work done by this force depends on the end points of the motion, not on the path in between.

WE0080

86. The alternative that gives the conservative force of the following is: **WE0223**

(A) $\vec{F}_1 = 2xy\hat{i} + x^2\hat{J}$ (B) $\vec{F}_2 = y^3\hat{i} + xy^2\hat{J}$ (C) $\vec{F}_3 = y\hat{i} + x\hat{J}$ (D) $\vec{F}_4 = xy^2\hat{i} + x^2\hat{J}$

Motion in a vertical circle

87. A particle is given a velocity at the bottom most position in a smooth spherical shell of radius 2m. It just complete a vertical circular motion. Then

(A) acceleration of particle when the velocity of the particle is in vertically upward direction is $g\sqrt{10}$

(B) acceleration of particle when the velocity of the particle is in vertically downward direction is $g\sqrt{10}$

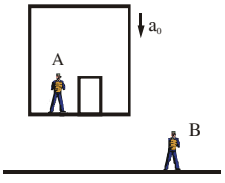
(C) acceleration of the particle at the top most point of path is g

(D) acceleration of the particle at the bottom most point is 5 g **WE0082**

COMPREHENSION TYPE QUESTIONS

Paragraph for Question no. 88 to 91

A block of mass m is kept in an elevator which starts moving downward with an acceleration a_0 ($a_0 < g$) as shown in figure. The block is observed by two observers A and B for a time interval t_0.



88. The observer B finds that the work done by gravity is

(A) $\frac{1}{2}mg^2t_0^2$ (B) $-\frac{1}{2}mg^2t_0^2$ (C) $\frac{1}{2}mga_0t_0^2$ (D) $-\frac{1}{2}mga_0t_0^2$

WE0083

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89. The observer B finds that work done by normal reaction N is :-

(A) zero (B) $-Na_0t_0^2$ (C) $\frac{Na_0t^2}{2}$ (D) None of these

WE0083

90. According to observer B, the net work done on the block is

(A) $-\frac{1}{2}ma_0t_0^2$ (B) $\frac{1}{2}ma_0^2t_0^2$ (C) $\frac{1}{2}mga_0t_0^2$ (D) $-\frac{1}{2}mga_0t_0^2$

WE0083

91. According to the observer A

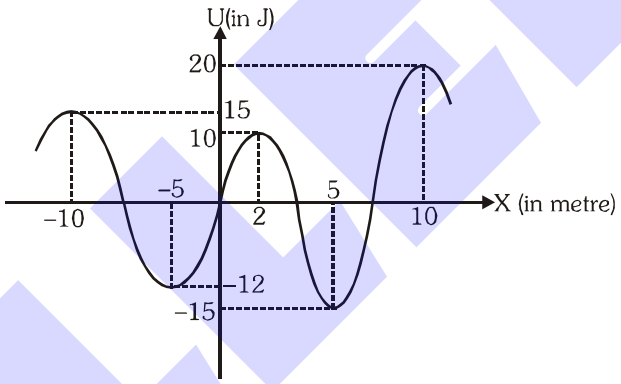
(A) the work done by gravity is zero (B) the work done by normal reaction is zero

(C) the work done by pseudo force is zero (D) all the above

WE0083

Paragraph for Question No. 92 to 94

In the figure the variation of potential energy of a particle of mass m = 2kg is represented w.r.t. its x-coordinate. The particle moves under the effect of this conservative force along the x-axis.



92. If the particle is released at the origin then :

(A) It will move towards positive x-axis.

(B) It will move towards negative x-axis.

(C) It will remain stationary at the origin.

(D) Its subsequent motion cannot be decided due to lack of information.

WE0070

93. If the particle is released at $x = 2 + \Delta$ where $\Delta \to 0$ (it is positive) then its maximum speed in subsequent motion will be-

(A) $\sqrt{22}$ m/s (B) $\sqrt{25}$ m/s (C) $\sqrt{24}$ m/s (D) $\sqrt{23}$ m/s

WE0070

94. x = –5 m and x = 10 m positions of the particle are respectively of-

(A) Neutral and stable equilibrium (B) Neutral and unstable equilibrium

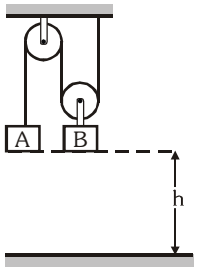
(C) Unstable and stable equilibrium (D) Stable and unstable equilibrium.

WE0070

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MATRIX MATCH TYPE QUESTIONS

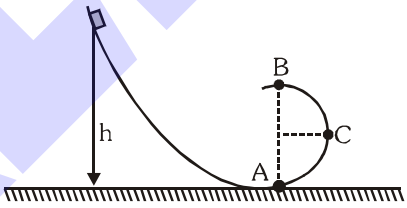
95. In the figure shown are two blocks A and B of same mass connected with pulley and string to each other. Initially both of them are at a height of h = 0.5 m from ground. After they are released they move in either direction and one of them strike the ground. For, the interval from releasing to when one of them strike, some physical quantities are in column I and their modulus values in SI units are in column II.



	Column I		**Column II**
(A)	Velocity of A immediately before any one of them strike ground.	(P)	1
(B)	Velocity of B immediately before any one of them strike ground.	(Q)	2
(C)	Ratio of work done by gravity on A to workdone by gravity on B.	(R)	3
(D)	Acceleration of block A before any one of them strike ground.	(S)	4
		(T)	5

WE0071

96. A block of mass m is released from top of a smooth track as shown in the figure. The end part of the track is a circle in vertical plane of radius R. N is normal reaction of the track at any point of the track. Match the entries of column I with entries of column–II.



	Column I		**Column II**
(A)	$h = \frac{5}{2}R$	(P)	Net force on the block at C is mg
(B)	$h = \frac{9}{2}R$	(Q)	$N_A - N_B = 6$ mg
(C)	h = R	(R)	Block leaves contact before B
(D)	h = 2R	(S)	Block will keep contact with the track in region between A & B.
		(T)	$N_C > mg$

WE0072

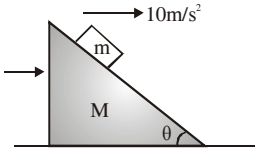
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EXERCISE (O-2)

SINGLE CORRECT TYPE QUESTIONS

Work energy theorem

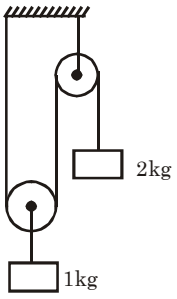
1. In the figure shown all the surfaces are frictionless, and mass of the block, m = 1 kg. The block and wedge are held initially at rest. Now wedge is given a horizontal acceleration of 10 m/s^2 by applying a force on the wedge, so that the block does not slip on the wedge. Then work done by the normal force in ground frame on the block in $\sqrt{3}$ seconds is :



(A) 30J (B) 60 J (C) 150 J (D) 100 $\sqrt{3}$ J

WE0041

2. In the pulley–block system shown in figure, strings are light. Pulleys are massless and smooth. System is released from rest. In 0.3 seconds



(a) work done on 2 kg block by gravity is 6J

(b) work done on 2 kg block by string is –2J

(c) work done on 1 kg block by gravity is –1.5J

(d) work done on 1 kg block by string is 2 J

(A) Only a, d are correct (B) Only b, d are correct

(C) Only a, b, c are correct (D) All are correct

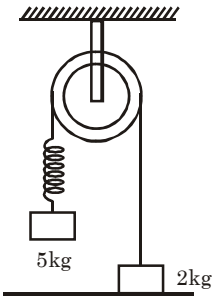
WE0224

3. A light spring of length 20 cm and force constant 2 N/cm is placed vertically on a table. A small block of mass 1 kg falls on it. The length h from the surface of the table at which the block will have the maximum velocity is :

(A) 20 cm (B) 15 cm (C) 10 cm (D) 5cm

WE0075

4. System shown in fig is released from rest with mass 2kg in contact with the ground. Pulley and spring are massless and the friction is absent everywhere. The speed of 5kg block when 2kg block leaves the contact with the ground is (force constant of the spring k = 40 N/m and g = 10m/s^2)

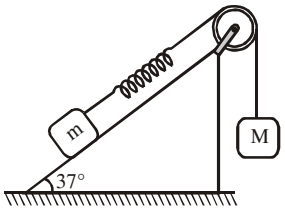


(A) $\sqrt{2}$ m/s (B) $2\sqrt{2}$ m/s (C) 2 m/s (D) $\sqrt{2}$ m/s

WE0185

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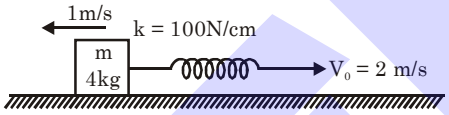
5. A block of mass m is attached with a massless spring of force constant k. The block is placed over a rough inclined surface for which the coefficient of friction is $\mu = \frac{3}{4}$. The minimum value of M required to move the block up the plane is : (Neglect mass of string and pulley and friction in pulley)



(A) $\frac{3}{5}$ m (B) $\frac{4}{5}$ m (C) 2 m (D) $\frac{3}{2}$ m

WE0052

6. The spring block system lies on a smooth horizontal surface. The free end of the spring is being pulled towards right with constant speed $v_0 = 2$m/s. At t = 0 sec, the spring of constant k = 100 N/ cm is unsttretched and the block has a speed 1 m/s to left. The maximum extension of the spring is:



(A) 2 cm (B) 4 cm (C) 6 cm (D) 8 cm **WE0225**

7. A particle moves in a straight line with retardation proportional to its displacement. Its loss of kinetic energy for any displacement x is proportional to- **[AIEEE - 2004]**

(A) x^2 (B) e^x (C) x (D) $\log_e x$

WE0054

8. A particle is projected along a horizontal surface whose coefficient of friction varies as $\mu = \frac{A}{r^2}$, where r is the distance from the origin in metres and A is a positive constant. The initial distance of the particle is 1m from the origin and its velocity is radially outwards. The minimum initial velocity at this point so the particle never stops is :

(A) ∞ (B) $2\sqrt{gA}$ (C) $\sqrt{2gA}$ (D) $4\sqrt{gA}$ **WE0226**

Potential energy

9. The potential energy of a 1 kg particle free to move along the x-axis is given by $U(x) = \left(\frac{x^4}{4} - \frac{x^2}{2}\right)$J. The total mechanical energy of the particle is 2J. Then, the maximum speed in (m/s) is:

(A) 1/ 2 (B) 2 (C) $\frac{3}{\sqrt{2}}$ (D) 2

WE0186

10. The potential energy of a particle is determined by the expression $U = a(x^2 + y^2)$, where a is a positive constant. The particle begins to move from a point with coordinates (3, 3), only under the action of potential field force. Then its kinetic energy T at the instant when the particle is at a point with the coordinates (1, 1) is :

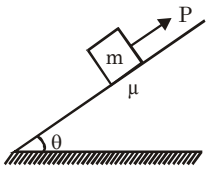
(A) 8a (B) 24a (C) 16a (D) Zero

WE0187

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Power

11. A block of mass m is being pulled up the rough inclined by an agent delivering constant power P. The coeficient of friction between the block and the inclined is μ. The maximum speed of the block during the course of ascent is:



(A) $v = \frac{P}{mg\sin\theta + \mu mg\cos\theta}$ (B) $v = \frac{P}{mg\sin\theta - \mu mg\cos\theta}$

(C) $v = \frac{2P}{mg\sin\theta - \mu mg\cos\theta}$ (D) $v = \frac{3P}{mg\sin\theta - \mu mg\cos\theta}$

WE0188

12. Assume the aerodynamic drag force on a car is proportional to its speed. If the power output from the engine is doubled, then the maximum speed of the car.

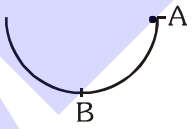
(A) is unchanged (B) increases by a factor of $\sqrt{2}$

(C) is also doubled (D) increases by a factor of four.

WE0056

Motion in a vertical circle

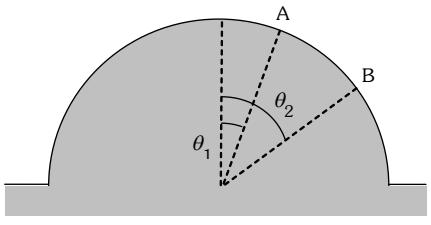
13. A small ball can move in a vertical plane along a semi–circle of radius r without friction. At what speed is the ball to launch from point A so that its acceleration is 3g at point B ?



(A) $(3gr)^{1/2}$ (B) $(2gr)^{1/2}$ (C) $(gr)^{1/2}$ (D) $2(gr)^{1/2}$

WE0076

14. A small block slides down from rest at point A on the surface of a smooth circular cylinder, as shown. At point B, the block falls off (leaves) the cylinder. The equation relating the angles θ_1 and θ_2 is given by



(A) $\sin\theta_2 = \frac{2}{3}\sin\theta_1$ (B) $\sin\theta_2 = \frac{3}{2}\sin\theta_1$ (C) $\cos\theta_2 = \frac{2}{3}\cos\theta_1$ (D) $\cos\theta_2 = \frac{3}{2}\cos\theta_1$

WE0066

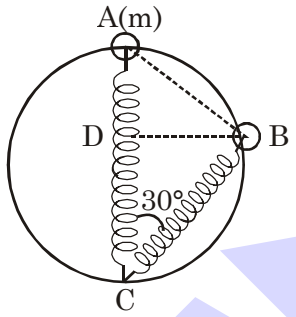
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15. A smooth sphere of radius R is made to translate in a straight line with a constant acceleration a = g. A particle kept on the top of the sphere is released from there at zero velocity with respect to the sphere. The speed of the particle with respect to sphere as a function of θ as it slides down is:

(A) $\sqrt{gR(1+\sin\theta-\cos\theta)}$ (B) $\sqrt{gR(1+\cos\theta-\sin\theta)}$

(C) $\sqrt{2gR(1-\cos\theta-\sin\theta)}$ (D) $\sqrt{2gR(1+\sin\theta-\cos\theta)}$ **WE0227**

16. A ring 'A' of mass 'm' is attached to a stretched spring of force constant K, which is fixed at C on a smooth vertical circular track of radius R. Points A and C are diametrically opposite. When the ring slips from rest on the track to point B, making an angle of 30° with AC. (∠ACB = 30°) spring becomes unstretched. Find the velocity of the ring at B.



(A) $\left[\frac{KR^2}{2m}(2-\sqrt{3})^2+gR\sqrt{3}\right]^{1/2}$ (B) $\left[\frac{KR^2}{m}(2-\sqrt{3})^2+gR\right]^{1/2}$

(C) $\left[\frac{2KR^2}{m}(2-\sqrt{3})^2+gR\sqrt{3}\right]^{1/2}$ (D) $\left[\frac{KR^2}{2m}(\sqrt{2}-1)^2+gR\right]^{1/2}$ **WE0228**

MULTIPLE CORRECT TYPE QUESTIONS

17. A particle of mass 2 kg is projected with an initial speed u = 10 m/sec at an angle θ = 30° with the horizontal

(A) Total work done on the particle during the first half of the total time of flight of the particle is (–25) J.

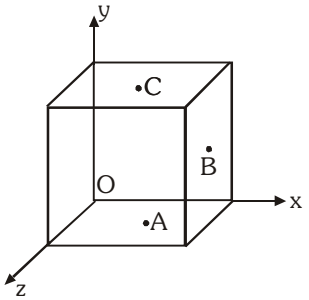
(B) Total work done on the particle during the total time of flight of the particle is 0 J.

(C) Average power delivered to the particle during the first half of the flight is (–50) watt.

(D) The radius of curvature of the trajectory of the particle at the highest point of the projectile is 7.5m.

WE0077

18. A particle is shifted from A to B and then from B to C where A, B and C are the midpoints of the corresponding faces of a cube of side 2m. If a force $\vec{F}=(3\hat{i}+4\hat{j}-5\hat{k})N$ is continuously acting on the particle, then select correct alternative :-



(A) work done from A to B is 7 J (B) work done from B to C is 1 J

(C) work done A to C is 8 J (D) $\vec{F}$ is a conservative force **WE0078**

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19. Which of the following is/are conservative force(s)?

(A) $\vec{F} = 2r^3\hat{r}$ (B) $\vec{F} = -\frac{5}{r}\hat{r}$ (C) $\vec{F} = \frac{3(x\,i + y\,\hat{j})}{(x^2+y^2)^{3/2}}$ (D) $\vec{F} = \frac{3(y\,i + x\,\hat{j})}{(x^2+y^2)^{3/2}}$

WE0079

20. A small sphere of mass m suspended by a thread is first taken a side so that the thread forms the right angle with the vertical and then released, then **WE0229**

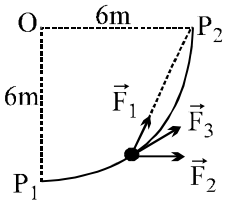
(A) Total acceleration of sphere as a function of θ measured from the vertical is $g\sqrt{1+3\cos^2\theta}$.

(B) Thread tension as a function of θ measured from the vertical is T = 3mg cosθ.

(C) The angle θ between the thread and the vertical at the moment when the total acceleration vector of the sphere is directed horizontally is $\cos^{-1}\frac{1}{\sqrt{3}}$.

(D) The thread tension at the moment when the vertical component of the sphere's velocity is maximum will be mg.

21. A smooth track in the form of a quarter circle of radius 6 m lies in the vertical plane. A particle moves from P_1 to P_2 under the action of forces $\vec{F}_1$, $\vec{F}_2$ and $\vec{F}_3$. Force $\vec{F}_1$ is always toward P_2 and is always 20N in magnitude. Force $\vec{F}_2$ always acts horizontally and is always 30 N in magnitude. Force $\vec{F}_3$ always acts tangentially to the track and is of magnitude 15 N. Select the correct alternative(s)



(A) work done by $\vec{F}_1$ is 120 J

(B) work done by $\vec{F}_2$ is 180 J

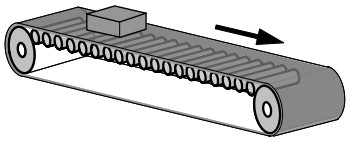
(C) work done by $\vec{F}_3$ is 45 π

(D) $\vec{F}_1$ is conservative in nature

COMPREHENSION TYPE QUESTIONS

Paragraph for Question 22 & 23

An object of mass M is gently placed on a horizontal conveyor belt, which is moving with uniform velocity v_o as shown in the figure. The coefficient of static friction is μ_s, the coefficient of kinetic friction is μ_k, and the acceleration of gravity is g. Initially the object slips for a while but finally moves without slipping together with the belt.



22. How far the conveyor belt moves while the object is slipping?

(A) $\frac{v_o^2}{\mu_k g}$ (B) $\frac{v_o^2}{2\mu_k g}$ (C) $\frac{v_o^2}{\mu_s g}$ (D) $\frac{v_o^2}{2\mu_s g}$

WE0084

23. Work done on the object by friction force relative to the reference frame moving with the conveyer belt is

(A) $\frac{1}{2}Mv_o^2$ (B) $-\frac{1}{2}Mv_o^2$ (C) Mv_o^2 (D) zero

WE0084

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EXERCISE (S)

Work

1. The sign of work done by a force on a body is important to understand. State carefully if the following quantities are positive or negative: **(NCERT)**

(a) work done by a man in lifting a bucket out of a well by means of a rope tied to the bucket.

(b) work done by gravitational force in the above case,

(c) work done by friction on a body sliding down an inclined plane,

(d) work done by an applied force on a body moving on a rough horizontal plane with uniform velocity,

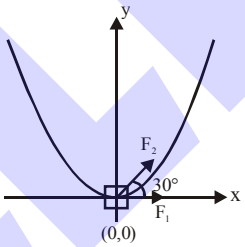
(e) work done by the resistive force of air on a vibrating pendulum in bringing it to rest.

WE0001

2. A point mass of 0.5 kg is moving along x-axis as $x = t^2 + 2t$, where, x is in meters and t is in seconds. Find the work done (in J) by all the forces acting on the body during the time interval [0, 2s].

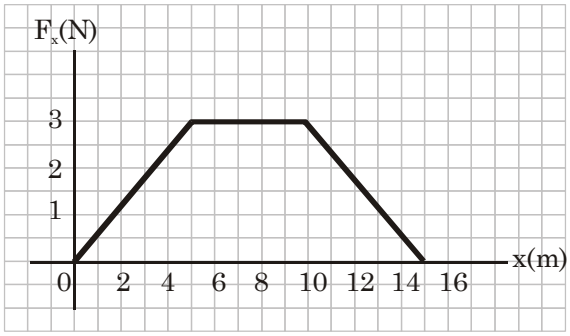
WE0003

3. A sleeve of mass 2 kg at origin can move on wire of parabolic shape $x^2 = 4y$. Two forces $F_1 = 6N$ and $F_2 = 8N$ are applied on the sleeve. F_1 is constant and is in x-direction. F_2 is constant in direction and magnitude. Body is displaced from origin to x = 4, then net work done by F_1 and F_2 is



WE0004

4. A particle is subject to a force F_x that varies with position as in figure. Find the work done by the force on the body as it moves (a) from x = 0 to x = 5.00 m, (b) from x = 5.00 m to x = 10.0 m, and (c) from x = 10.0 m to x = 15.0 m. (d) What is the total work done by the force over the distance x = 0 to x = 15.0 m?



WE0005

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5. A spring, which is initially in its unstretched condition, is first stretched by a length x and then again by a further length x. The work done in the first case is W_1 and in the second case is W_2. Find $\frac{W_2}{W_1}$.

WE0006

Kinetic energy, Work energy theorem

6. A body of mass 2 kg initially at rest moves under the action of an applied horizontal force of 7 N on a table with coefficient of kinetic friction = 0.1. Compute the **(NCERT)**

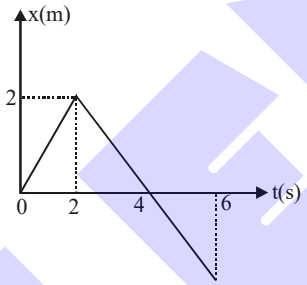
(a) work done by the applied force in 10 s,

(b) work done by friction in 10 s,

(c) work done by the net force on the body in 10 s,

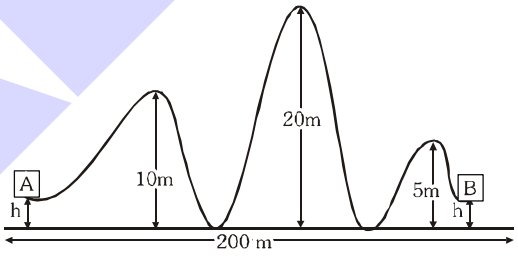
(d) change in kinetic energy of the body in 10 s, and interpret your results. **WE0007**

7. Position-time graph of a particle of mass 2 kg is shown in figure. Total work done on the particle from t = 0 to t = 4s is



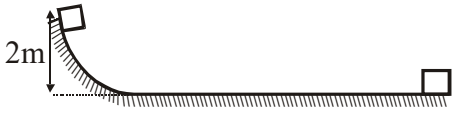
WE0008

8. A point object of mass 2 kg is moved from point A to point B very slowly on a curved path by applying a tangential force on a curved path as shown in figure. Then find the work done by external force in moving the body. Given that $\mu_s = 0.3$, $\mu_k = 0.1$. [$g = 10\ m/s^2$]



WE0009

9. A block is released from rest from top of a rough curved track as shown in figure. It comes to rest at some point on the horizontal part. If its mass is 200 gm, calculate negative of work by friction in joules.

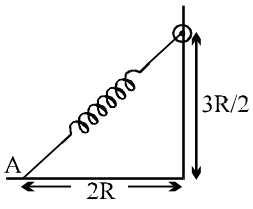


WE0011

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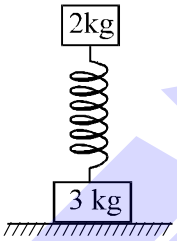
10. A ring of mass m can slide over a smooth vertical rod. The ring is connected to a spring of force constant $K = \frac{4mg}{R}$ where 2R is the natural length of the spring.



The other end of the spring is fixed to the ground at a horizontal distance 2R from the base of the rod. The mass is released at a height of 1.5R from ground
(i) calculate the work done by the spring.
(ii) calculate the velocity of the ring as it reaches the ground.

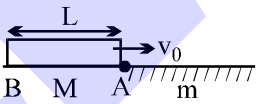
WE0027

11. The ends of spring are attached to blocks of mass 3kg and 2kg. The 3kg block rests on a horizontal surface and the 2kg block which is vertically above it is in equilibrium producing a compression of 1cm of the spring. The 2kg mass must be compressed further by at least _____, so that when it is released, the 3 kg block may be lifted off the ground.



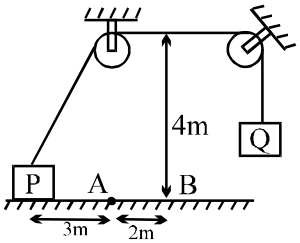
WE0028

12. A uniform rod of mass *m* length *L* is sliding along its length on a horizontal table whose top is partly smooth & rest rough with friction coefficient μ. If the rod after moving through smooth part, enters the rough with velocity v_0.



(i) What will be the magnitude of the friction force when its x length (< L) lies in the rough part during sliding.
(ii) Determine the minimum velocity v_0 with which it must enter so that it lies completely in rough region before coming to rest.
(iii) If the velocity is double the minimum velocity as calculated in part (a) then what distance does its front end A would have travelled in rough region before rod comes to rest. **WE0029**

13. Two blocks P and Q of mass 2 m and m respectively are connected by a massless string and are at rest as shown in fig. all pulleys are ideal and the surface is frictionless. Find the velocity of the block P at point A and B when the system is released from rest. [at A, thread from P to pulley is vertical]



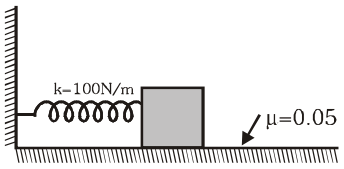
WE0230

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14. A particle is moving with kinetic energy E, straight up an inclined plane with angle α, the coefficient of friction being μ. The work done against friction before the particle comes down to rest is

WE0033

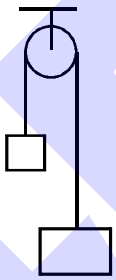
15. A block of mass 1 kg is attached to a spring with a force constant 100 N/m and rests on a rough horizontal ground as shown in the figure. Initial displacement of block from natural length is 50 cm. The total distance covered by the block if coefficient of friction between block & ground is 0.05. [g =10m/s^2]



WE0034

16. A light inextensible string that goes over a smooth fixed pulley as shown in the figure connects two blocks of masses 0.36 kg and 0.72 kg. Taking g = 10 m/s^2, find the work done (in **joules)** by the string on the block of mass 0.36 kg during the first second after the system is released from rest.

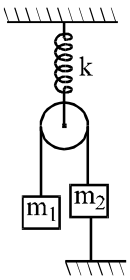
[IIT-JEE-2009]



WE0096

Potential Energy

17. In the figure shown, pulley and spring are ideal. Find the potential energy stored in the spring ($m_1 > m_2$).



WE0017

18. The potential energy (in joules) function of a particle in a region of space is given as :

$$U = (2x^2 + 3y^3 + 2z)$$

Here x, y and z are in metres. Find the magnitude of x component of force (in newton) acting on the particle at point P (1m, 2m, 3m).

WE0018

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19. The P.E. of a particle oscillating on x-axis is given as $U = 20 + (x - 2)^2$ here U is in Joules & x is in meters. Total mechanical energy of particle is 36 J

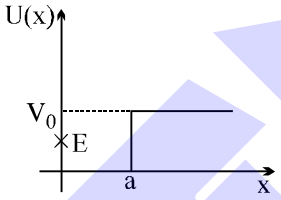
(i) Find the mean position (ii) Find the max. K.E. of the particle

WE0019

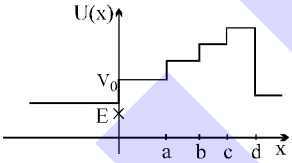
20. The potential energy of a 2 kg particle moving along the x axis is given by $U(x) = (4.0J/m^2)x^2 + (1.0J/m^4)x^4$. When the particle is at x = 1.0m, find its acceleration. [only conservative forces are acting]

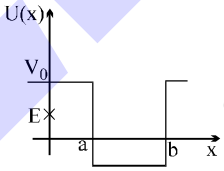
WE0020

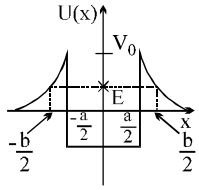
21. The given graph is a potential energy function in one dimension. The total energy of particle is indicated by cross on the ordinate axis. The graph of figure-1 is given as an example.From the figure-1, it can be interpreted that for the given total energy indicated by cross on the ordinate axis the particle cannot be found in the Region : x > a. Now, for the following potential functions in one dimensions, specify the regions, in which the particle cannot be found for the energy marked as E on graphs. Give your answer in the blocks shown.



(Figure -1)

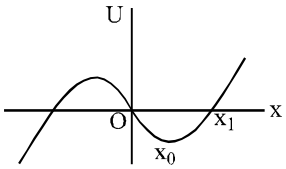
(Graph-1) 

(Graph-2) 

(Graph-3) 

WE0023

22. A particle is confined to move along the +x axis under the action of a force F(x) that is derivable from the potential energy $U(x) = ax^3 - bx$.



(i) Find the expression for F(x)

(ii) When the total energy of the particle is zero, the particle can be trapped with in the interval x=0 to $x = x_1$. For this case find the values of x_1.

(iii) Determine the maximum kinetic energy that the trapped particle has in its motion. Express all answers in terms a and b. At what value of x will the kinetic energy be maximum ?

WE0030

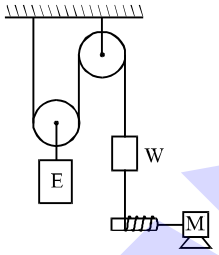
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Power

23. A 4 kg particle moves along the X-axis. It's position x varies with time according to $x(t) = t + 2t^3$, where x is in m and t is in seconds. Compute:
(i) The kinetic energy at time t.
(ii) The force acting on the particle at time t.
(iii) The power delivered to the particle at time t.
(iv) The work done on the particle from t = 0 to t = 2 seconds.

WE0010

24. The elevator E has a mass of 3000 kg when fully loaded and is connected as shown to a counterweight W of mass 1000 kg. Determine the power in kilowatts delivered by the motor
(a) when the elevator is moving down at a constant speed of 3 m/s,
(b) when it has an upward velocity of 3 m/s and a deceleration of 0.5 m/s^2.



WE0015

25. Power applied to a particle varies with time as $P = (3t^2 - 2t + 1)$ watt, where t is in second. Find the change in its kinetic energy between time t = 2 s and t = 4 s.

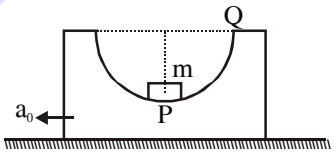
WE0016

26. A 650-kg elevator starts from rest. It moves upward for 3.00 s with constant acceleration until it reaches its cruising speed of 1.75 m/s.
(i) What is the average power of the elevator motor during this period?
(ii) How does this power compare with its power when it moves at its cruising speed?

WE0032

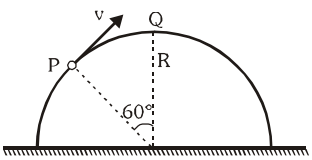
Motion in vertical circle

27. A small block of mass m is lying at rest at point P of a wedge having a smooth semi circular track of radius R. The minimum value of horizontal acceleration a_0 (in m/s^2) of wedge so that mass can just reach the point Q, is



WE0013

28. A particle is given a certain veloicty v at point P as shown on a hemispherical smooth surface. Find the value of v (in m/s), such that when particle reaches Q, the normal reaction of surface becomes equal to particle's weight. [R = 1.6 m, g = 10 m/s^2]

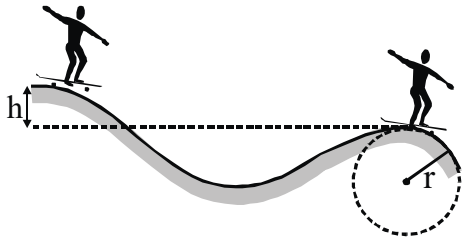


WE0021

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29. A skier starts from rest at the top of a hill. The skier coasts down the hill and up a second hill, as the drawing illustrates. The crest of the second hill is circular, with a radius of r = 36 m. Neglect friction and air resistance. What must be the height h (in m) of the first hill so that the skier just loses contact with the snow at the crest of the second hill?

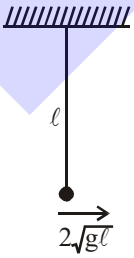


WE0022

30. A ball is attached to a horizontal cord of length L whose other end is fixed, (a) If the ball is released, what will be its speed at the lowest point of its path ? (b) A peg is located a distance h directly below the point of attachment of the cord. If h = 0.75 L, what will be the speed of the ball when it reaches the top of its circular path about the peg ?

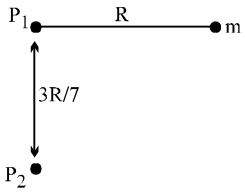
WE0025

31. One end of a string of length $\ell = \frac{14}{9}$ m is fixed and a mass of 1 kg is tied to the other end. The ball is given a velocity $2\sqrt{g\ell}$ at the bottom most point as shown in figure. The string is cut when the ball becomes horizontal. Find the distance (in m) travelled till it stop for the 1st time (Take $\pi = \frac{22}{7}$).



WE0026

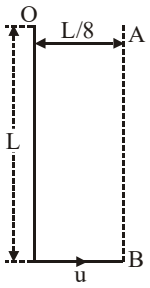
32. A simple pendulum consists of a bob of mass m and a string of length R suspended from a peg P_1 on the wall. A second peg P_2 is fixed vertically below the first one at a distance $3R/7$ from it. The pendulum is drawn aside such that the string is horizontal and released. Calculate the maximum height (with respect to the lowest point) to which it rises



WE0035

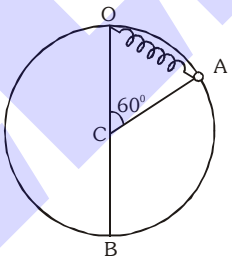
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33. A particle is suspended vertically from a point O by an inextensible massless string of length L. A vertical line AB is at a distance $\frac{L}{8}$ from O as shown in figure. The object is given a horizontal velocity u. At some point, its motion ceases to be circular and eventually the object passes through the line AB. At the instant of crossing AB, its velocity is horizontal. Find u.



WE0036

34. A particle of mass 5 kg is free to slide on a smooth ring of radius r = 20 cm fixed in a vertical plane. The particle is attached to one end of a spring whose other end is fixed to the top point O of the ring. Initially the particle is at rest at a point A of the ring such that $\angle$ OCA = 60^0, C being the centre of the ring. The natural length of the spring is also equal to r = 20 cm. After the particle is released and slides down the ring the contact force between the particle & the ring becomes zero when it reaches the lowest position B. Determine the force constant of the spring.



WE0037

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EXERCISE (JM)

1. When a rubber-band is stretched by a distance x, it exerts a restoring force of magnitude $F = ax + bx^2$ where a and b are constants. The work done in stretching the unstretched rubber-band by L is:- **[JEE-Main-2014]**

(1) $\frac{aL^2}{2}+\frac{bL^3}{3}$ (2) $\frac{1}{2}\left(\frac{aL^2}{2}+\frac{bL^3}{3}\right)$ (3) $aL^2 + bL^3$ (4) $\frac{1}{2}(aL^2+bL^3)$

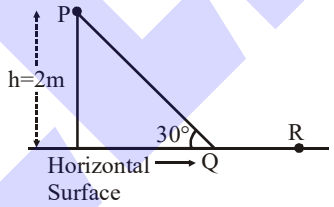
WE0090

2. A person trying to lose weight by burning fat lifts a mass of 10 kg upto a height of 1 m 1000 times. Assume that the potential energy lost each time he lowers the mass is dissipated. How much fat will he use up considering the work done only when the weight is lifted up? Fat supplies 3.8×10^7 J of energy per kg which is converted to mechanical energy with a 20% efficiency rate. Take $g = 9.8\ ms^{-2}$:- **[JEE-Main-2016]**

(1) 12.89×10^{-3} kg (2) 2.45×10^{-3} kg (3) 6.45×10^{-3} kg (4) 9.89×10^{-3} kg

WE0091

3. A point particle of mass, moves along the uniformly rough track PQR as shown in the figure. The coefficient of friction, between the particle and the rough track equals μ. The particle is released, from rest, from the point P and it comes to rest at a point R. The energies, lost by the ball, over the parts, PQ and PR, of the track, are equal to each other, and no energy is lost when particle changes direction from PQ to QR. The values of the coefficient of friction μ and the distance x(=QR) are, respecitvely close to :- **[JEE-Main-2016]**



(1) 0.29 and 6.5 m (2) 0.2 and 6.5 m (3) 0.2 and 3.5 m (4) 0.29 and 3.5 m

WE0092

4. A body of mass $m = 10^{-2}$ kg is moving in a medium and experiences a frictional force $F = -kv^2$. Its intial speed is $v_0 = 10\ ms^{-1}$. If, after 10 s, its energy is $\frac{1}{8}mv_0^2$, the value of k will be :- **[JEE-Main-2017]**

(1) 10^{-4} kg m^{-1} (2) 10^{-1} kg m^{-1} s^{-1} (3) 10^{-3} kg m^{-1} (4) 10^{-3} kg s^{-1}

WE0093

5. A time dependent force F = 6t acts on a particle of mass 1 kg. If the particle starts from rest, the work done by the force during the first 1 sec. will be : **[JEE-Main-2017]**

(1) 9 J (2) 18 J (3) 4.5 J (4) 22 J

WE0094

6. A particle is moving in a circular path of radius a under the action of an attractive potential $U = -\frac{k}{2r^2}$. Its total energy is :- **[JEE-Main-2018]**

(1) $\frac{k}{2a^2}$ (2) Zero (3) $-\frac{3}{2}\frac{k}{a^2}$ (4) $-\frac{k}{4a^2}$

WE0095

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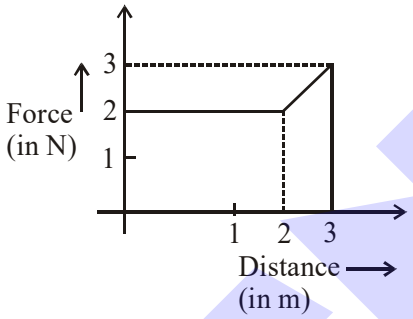
SELECTED PROBLEMS FROM JEE-MAINS ONLINE PAPERS

7. A particle which is experiencing a force, given by $\vec{F} = 3\hat{i} - 12\hat{j}$, undergoes a displacement of $\vec{d} = 4\hat{i}$. If the particle had a kinetic energy of 3 J at the beginning of the displacement, what is its kinetic energy at the end of the displacement ? **[JEE-Main-2019_Jan.]**

(1) 15 J (2) 10 J (3) 12 J (4) 9 J

WE0140

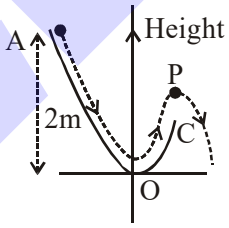
8. A particle moves in one dimension from rest under the influence of a force that varies with the distance travelled by the particle as shown in the figure. The kinetic energy of the particle after it has travelled 3m is : **[JEE-Main-2019_April]**



(1) 6.5 J (2) 2.5 J (3) 4 J (4) 5 J

WE0141

9. A particle (m = 1 kg) slides down a frictionless track (AOC) starting from rest at a point A (height 2 m). After reaching C, the particle continues to move freely in air as a projectile. When it reaching its highest point P (height 1 m), the kinetic energy of the particle (in J) is : (Figure drawn is schematic and not to scale; take g=10 ms^{-2})_____. **[JEE-Main-2020_Jan.]**



WE0142

10. An elevator in a building can carry a maximum of 10 persons, with the average mass of each person being 68 kg. The mass of the elevator itself is 920 kg and it moves with a constant speed 3 m/s. The frictional force opposing the motion is 6000 N. If the elevator is moving up with its full capacity, the power delivered by the motor to the elevator (g = 10 m/s^2) must be at least : **[JEE-Main-2020_Jan]**

(1) 56300 W (2) 48000 W (3) 66000 W (4) 62360 W

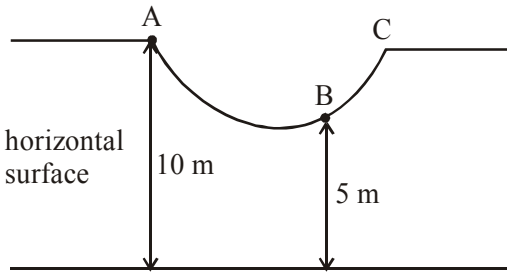
WE0143

11. A small bob tied at one end of a thin string of length 1m is describing a vertical circle so that the maximum and minimum tension in the string are in the ratio 5 : 1. The velocity of the bob at the height position is _____ m/s. (Take g = 10 m/s^2) **[JEE-Main-2021_Feb.]**

WE0144

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12. As shown in the figure, a particle of mass 10 kg is placed at a point A. When the particle is slightly displaced to its right, it starts moving and reaches the point B. The speed of the particle at B is x m/s. (Take g = 10 m/s^2). The value of 'x' to the nearest integer is_____. **[JEE-Main-2021_March]**

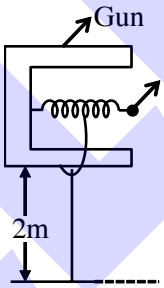


WE0145

13. A ball of mass 4 kg, moving with a velocity of 10 ms^{-1}, collides with a spring of length 8 m and force constant 100 Nm^{-1}. The length of the compressed spring is x m. The value of x, to the nearest integer, is_____. **[JEE-Main-2021_March]**

WE0146

14. In a spring gun having spring constant 100 N/m a small ball 'B' of mass 100 g is put in its barrel (as shown in figure) by compressing the spring through 0.05 m. There should be a box placed at a distance 'd' on the ground so that the ball falls in it. If the ball leaves the gun horizontally at a height of 2 m above the ground. The value of d is ____ m. (g = 10 m/s^2) **[JEE-Main-2021_July]**



WE0147

15. If the Kinetic energy of a moving body becomes four times its initial Kinetic energy, then the percentage change in its momentum will be : **[JEE-Main-2021_July]**

(1) 100% (2) 200% (3) 300% (4) 400%

WE0148

16. A porter lifts a heavy suitcase of mass 80 kg and at the destination lowers it down by a distance of 80 cm with a constant velocity. Calculate the workdone by the porter in lowering the suitcase. (take g = 9.8 ms^{-2}) **[JEE-Main-2021_July]**

(1) -62720.0 J (2) –627.2 J (3) +627.2 J (4) 784.0 J

WE0149

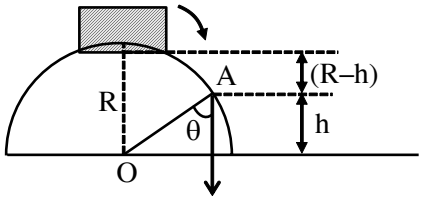
17. A boy is rolling a 0.5 kg ball on the frictionless floor with the speed of 20 ms^{-1}. The ball gets deflected by an obstacle on the way. After deflection it moves with 5% of its initial kinetic energy. What is the speed of the ball now ? **[JEE-Main-2021_July]**

(1) 19.0 ms^{-1} (2) 4.47 ms^{-1} (3) 14.41 ms^{-1} (4) 1.00 ms^{-1}

WE0150

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18. A small block slides down from the top of hemisphere of radius R = 3 m as shown in the figure. The height 'h' at which the block will lose contact with the surface of the sphere is ____m. (Assume there is no friction between the block and the hemisphere) **[JEE-Main-2021_Aug.]**



WE0151

19. Two persons A and B perform same amount of work in moving a body through a certain distance d with application of forces acting at angle 45° and 60° with the direction of displacement respectively. The ratio of force applied by person A to the force applied by person B is $\frac{1}{\sqrt{x}}$. The value of x is ..... .

[JEE-Main-2021_Aug.]

WE0152

20. A body of mass 'm' dropped from a height 'h' reaches the ground with a speed of $0.8\sqrt{gh}$. The value of workdone by the air-friction is : **[JEE-Main-2021_Aug.]**

(1) –0.68 mgh (2) mgh (3) 1.64 mgh (4) 0.64 mgh

WE0153

21. A particle experiences a variable force $\vec{F} = (4x\hat{i} + 3y^2\hat{j})$ in a horizontal x-y plane. Assume distance in meters and force is newton. If the particle moves from point (1, 2) to point (2, 3) in the x-y plane, the Kinetic Energy changes by : **[JEE-Main-2022_June]**

(1) 50.0 J (2) 12.5 J (3) 25.0 J (4) 0 J

WE0189

22. A stone tide to a string of length L is whirled in a vertical circle with the other end of the string at the centre. At a certain instant of time, the stone is at its lowest position and has a speed u. The magnitude of change in its velocity, as it reaches a position where the string is horizontal, is $\sqrt{x(u^2 - gL)}$. The value of x is : **[JEE-Main-2022_June]**

(1) 3 (2) 2 (3) 1 (4) 5

WE0190

23. Sand is being dropped from a stationary dropper at a rate of 0.5 kgs^{-1} on a conveyor belt moving with a velocity of 5 ms^{-1}. The power needed to keep belt moving with the same velocity will be : **[JEE-Main-2022_July]**

(1) 1.25 W (2) 2.5 W (3) 6.25 W (4) 12.5 W

WE0191

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EXERCISE (JA)

1. The work done on a particle of mass m by a force, $K\left[\frac{x}{\left(x^2+y^2\right)^{3/2}}\hat{i}+\frac{y}{\left(x^2+y^2\right)^{3/2}}\hat{j}\right]$ (K being a constant of appropriate dimensions), when the particle is taken from the point (a, 0) to the point (0, a) along a circular path of radius a about the origin in the x-y plane is :- **[JEE-Advance-2013]**

(A) $\frac{2K\pi}{a}$ (B) $\frac{K\pi}{a}$ (C) $\frac{K\pi}{2a}$ (D) 0

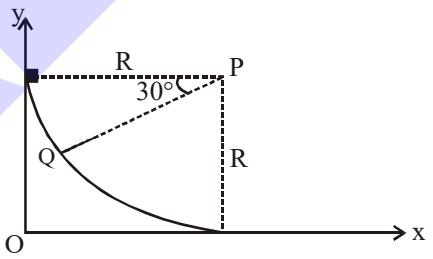
WE0098

2. A particle of mass 0.2 kg is moving in one dimension under a force that delivers a constant power 0.5W to the particle. If the initial speed (in ms^{-1}) of the particle is zero, the speed (in ms^{-1}) after 5 s is. **[JEE-Advance-2013]**

WE0099

Paragraph for Questions 3 and 4

A small block of mass 1 kg is released from rest at the top of a rough track. The track is a circular arc of radius 40 m. The block slides along the track without toppling and a frictional force acts on it in the direction opposite to the instantaneous velocity. The work done in overcoming the friction up to the point Q, as shown in the figure below, is 150 J. (Take the acceleration due to gravity, $g = 10\ m\ s^{-2}$) **[JEE-Advance-2013]**



3. The magnitude of the normal reaction that acts on the block at the point Q is

(A) 7.5 N (B) 8.6 N (C) 11.5 N (D) 22.5 N

WE0100

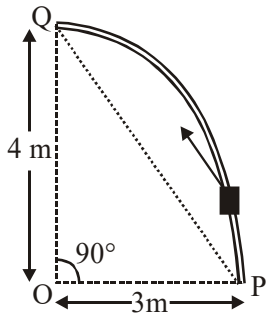
4. The speed of the block when it reaches the point Q is

(A) 5 ms^{-1} (B) 10 ms^{-1} (C) $10\sqrt{3} ms^{-1}$ (D) 20 ms^{-1}

WE0100

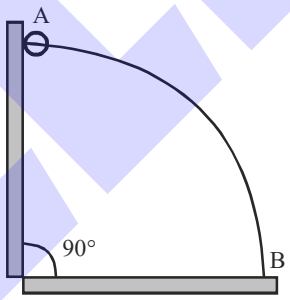
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5. Consider an elliptically shaped rail PQ in the vertical plane with OP = 3 m and OQ = 4m. A block of mass 1 kg is pulled along the rail from P to Q with a force of 18 N, which is always parallel to line PQ (see the figure given). Assuming no frictional losses, the kinetic energy of the block when it reaches Q is (n × 10) Joules. The value of n is (take acceleration due to gravity = 10 ms^{-2}) **[JEE-Advance-2014]**



WE0101

6. A wire, which passes through the hole in a small bead, is bent in the form of quarter of a circle. The wire is fixed vertically on ground as shown in the figure. The bead is released from near the top of the wire and it slides along the wire without friction. As the bead moves from A to B, the force it applies on the wire is :- **[JEE Advanced-2014]**

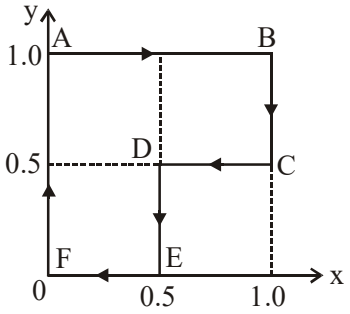


(A) Always radially outwards
(B) Always radially inwards
(C) Radially outwards initially and radially inwards later.
(D) Radially inwards initially and radially outwards later. **WE0102**

7. A particle of mass m is initially at rest at the origin. It is subjected to a force and starts moving along the x-axis. Its kinetic energy K changes with time as dK/dt = γt, where γ is a positive constant of appropriate dimensions. Which of the following statements is (are) true ? **[JEE Advanced-2018]**
(A) The force applied on the particle is constant
(B) The speed of the particle is proportional to time
(C) The distance of the particle from the origin increses linerarly with time
(D) The force is conservative **WE0103**

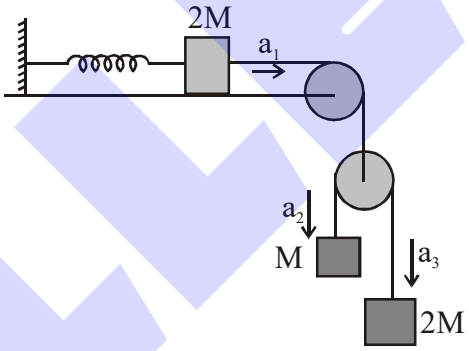
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8. A particle is moved along a path AB-BC-CD-DE-EF-FA, as shown in figure, in presence of a force $\vec{F} = \left(\alpha y\hat{i} + 2\alpha x\hat{j}\right)N$, where x and y are in meter and $\alpha = -1$ N/m^{-1}. The work done on the particle by this force $\vec{F}$ will be ____ Joule. **[JEE Advanced-2019]**



WE0104

9. A block of mass 2M is attached to a massless spring with spring-constant k. This block is connected to two other blocks of masses M and 2M using two massless pulleys and strings. The accelerations of the blocks are a_1, a_2 and a_3 as shown in figure. The system is released from rest with the spring in its unstretched state. The maximum extension of the spring is x_0. Which of the following option(s) is/are correct ? [g is the acceleration due to gravity. Neglect friction] **[JEE Advanced-2019]**



(1) $x_0 = \frac{4Mg}{k}$

(2) When spring achieves an extension of $\frac{x_0}{2}$ for the first time, the speed of the block connected to the spring is $3g\sqrt{\frac{M}{5k}}$

(3) $a_2 - a_1 = a_1 - a_3$

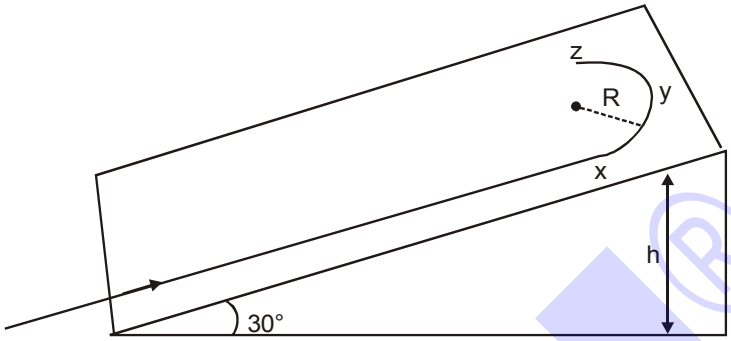
(4) At an extension of $\frac{x_0}{4}$ of the spring, the magnitude of acceleration of the block connected to the spring is $\frac{3g}{10}$

WE0105

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10. A student skates up a ramp that makes an angle 30° with the horizontal. He/she starts (as shown in the figure) at the bottom of the ramp with speed v_0 and wants to turn around over a semicircular path xyz of radius R during which he/she reaches a maximum height h (at point y) from the ground as shown in the figure. Assume that the energy loss is negligible and the force required for this turn at the highest point is provided by his/her weight only. Then (g is the acceleration due to gravity)

[IIT-JEE Advanced-2020]



(A) $v_0^2 - 2gh = \frac{1}{2}gR$

(B) $v_0^2 - 2gh = \frac{\sqrt{3}}{2}gR$

(C) the centripetal force required at points x and z is zero

(D) the centripetal force required is maximum at points x and z

CR0110

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ANSWER KEY

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	B	C	A	D	B	A	B	A	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	A	B	D	A	D	C	D	D	C
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	A	D	A	B	C	D	D	C	D	A
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	D	B	B	D	B	A	B	B	B	B
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	A	B	A	D	A	A	A	B	C	C
Que.	51	52	53	54	55	56	57	58	59	60
Ans.	D	C	D	A	A	D	D	A	A	A
Que.	61	62	63	64	65	66	67	68	69	70
Ans.	C	D	B	A	C	A	B	C	D	C
Que.	71	72	73	74	75	76	77	78	79	80
Ans.	C	A	B	C	A	A	A	C	D	A
Que.	81	82	83	84	85	86	87	88	89	90
Ans.	B,C,D	A,D	A,C	A,C	A,C,D	A,C	A,B,C, D	C	D	B
Que.	91	92	93	94	95	96				
Ans.	D	B	B	D	A-Q; B-P; C-Q; D-S	A-Q,S,T; B-Q,S,T; C-P,S; D-R,T				

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	D	B	B	A	C	A	C	C	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A	B	C	C	D	B	A,B,C,D	A,B,C,D	A,B,C	A,B,C
Que.	21	22	23							
Ans.	B,C,D	A	B							

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EXERCISE # S

1. (a) +ve; (b) –ve (c) –ve (d) +ve (e) –ve **2.** 8 **3.** 67.7 J **4.** (a) 7.5 J; (b) 15 J; (c) 7.5 J; (d) 30 J
5. 3 **6.** (a) 875 J; (b) –250 J; (c) 625 J ; (d) 625 J; **7.** 0 **8.** 400 J **9.** 4

10. (i) mgR/2, (ii) $2\sqrt{gR}$ **11.** 2.5cm **12.** (i) $f = \frac{\mu m}{\ell} xg$; (ii) $\sqrt{\mu g \ell}$; (iii) $\frac{5\ell}{2}$

13. $10^{1/2}$ m/s, $\sqrt{\frac{10g\left(5-2\sqrt{5}\right)}{11}}$ m/s **14.** $\frac{E\mu\cos\alpha}{\sin\alpha+\mu\cos\alpha}$ **15.** 25 m **16.** 8

17. $\frac{2m_1^2g^2}{k}$ **18.** 4 **19.** (i) x =2, (ii) 16 J **20.** –6 m/s^2

21. Graph-1 : For all x, Graph-2 : x < a & x > b, Graph-3 : $-\frac{b}{2} < x < -\frac{a}{2}$ & $\frac{a}{2} < x < \frac{b}{2}$

22. $F = -3ax^2 + b$, $x = \sqrt{\frac{b}{a}}$, $K_{max} = \frac{2b}{3}\sqrt{\frac{b}{3a}}$

23. (i) $2 + 24t^2 + 72t^4$ J, (ii) 48 t N, (iii) $48t + 288t^3$ W, (iv) 1248 J **24.** (a) –30 kW, 19.5 kW
25. 46 J **26.** (i) 6×10^3 W (ii) 1×10^4 W **27.** 10 **28.** 4

29. 018 **30.** $\sqrt{2gL}$, $\sqrt{gL}$ **31.** 4 **32.** 27R/28 **33.** $u = \sqrt{gL\left(2+\frac{3\sqrt{3}}{2}\right)}$ **34.** 500 N/m

EXERCISE # JM

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	1	4	1	3	2	1	1	10	3
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	5	10	6	1	1	2	2	2	2	1
Que.	21	22	23							
Ans.	3	2	4							

EXERCISE # JA

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	5	A	B	5	D	A,B,D	0.75	3	A,D

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